STAFF REVIEW · H446 290 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A distinct random per-account value, improving resilience of password hashing.

2. Matching records according to a join condition.

3. Page table.

4. 1+0+0=1=01₂; sum 1, carry 0.

I Do — source worked steps

1. Predict first — A page fault is a memory-management event where required virtual page data is not currently resident.

2. Show method — HTTP 404 is a web server status about a requested resource; the word 'page' refers to different concepts.

We Do — guided CHECK

1. Predict first
The input has no repeated runs longer than one symbol.
Show method
Encoding value/count pairs may consume more bits than the original; compression success depends on data redundancy.

You Do — source mark schemes

1. DNS resolves names to addresses [2]; ROM non-volatile [2]; hash is one-way digest [2]; INNER JOIN returns matched rows [2].

2. Cloud storage = remote files over network [3]; virtual memory = OS logical address mapping to frames/backing store [3].

3. Choose any three of the four misconceptions in Question 1. Award for each: a relevant, concrete counterexample [1] and an accurate mechanism-based explanation [1], maximum 6. Examples: (1) DNS resolves school.example to an IP address, while the web server sends HTML in an HTTP response; (2) UEFI firmware in non-volatile ROM/flash remains stored when the computer is powered off; (3) comparing a SHA-256 digest of a downloaded file with a trusted digest can detect changes, but no decryption of the digest recovers the file; (4) an INNER JOIN of Student and Score excludes a pupil with no matching Score row. Accept other correct examples.

Exit Ticket — source answer

1. False. Apply it to an unseen example.